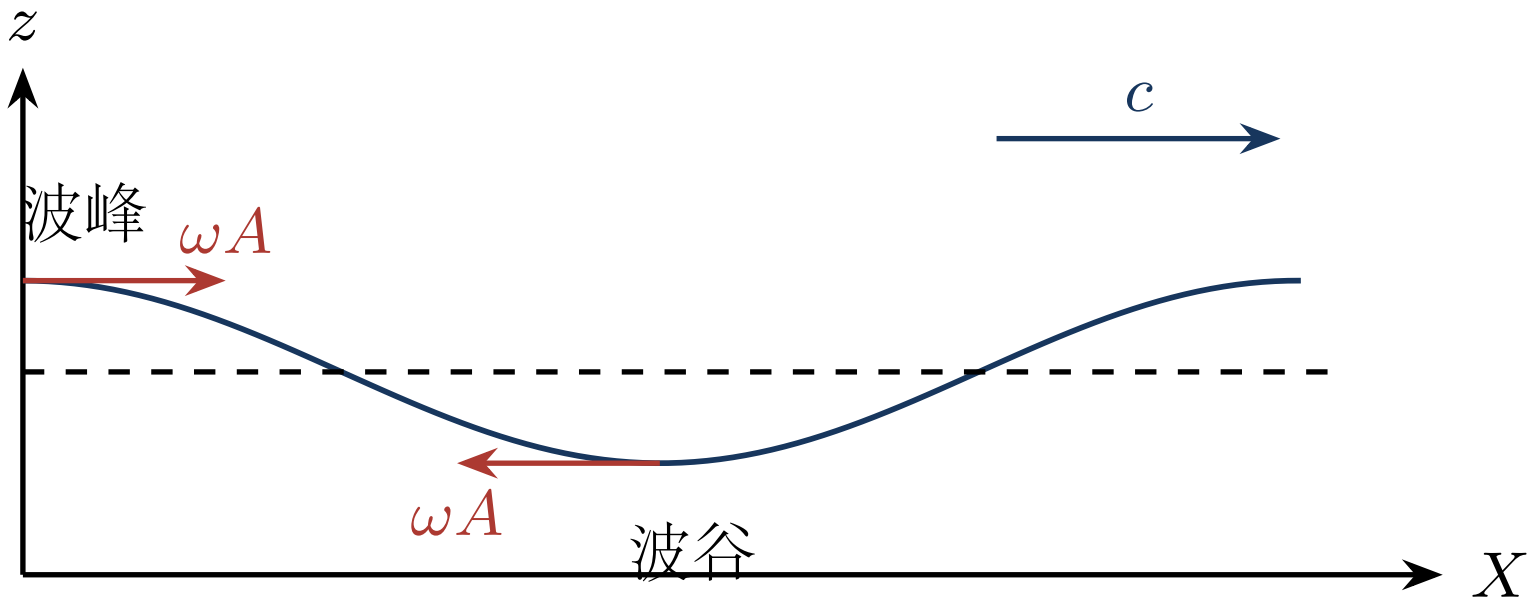
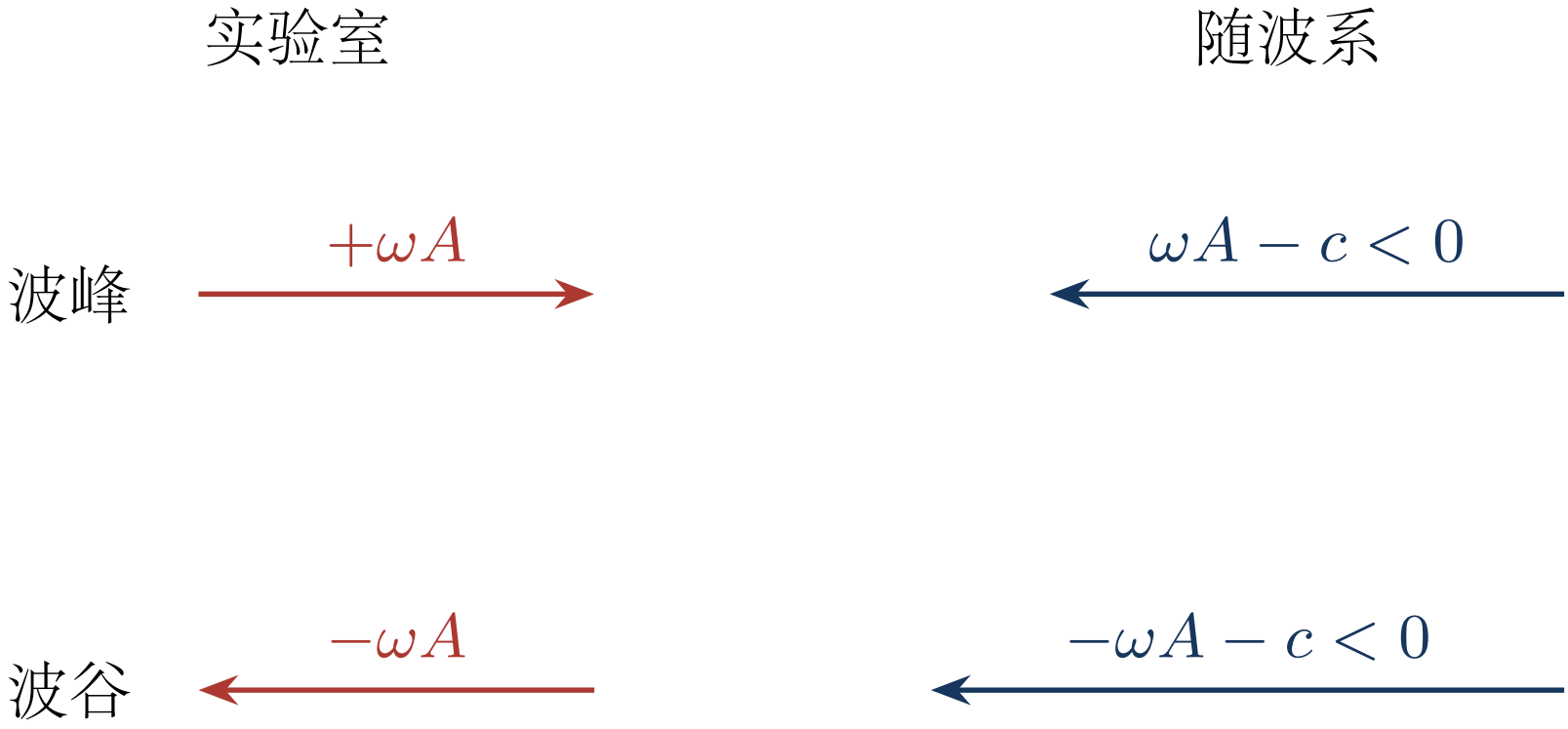
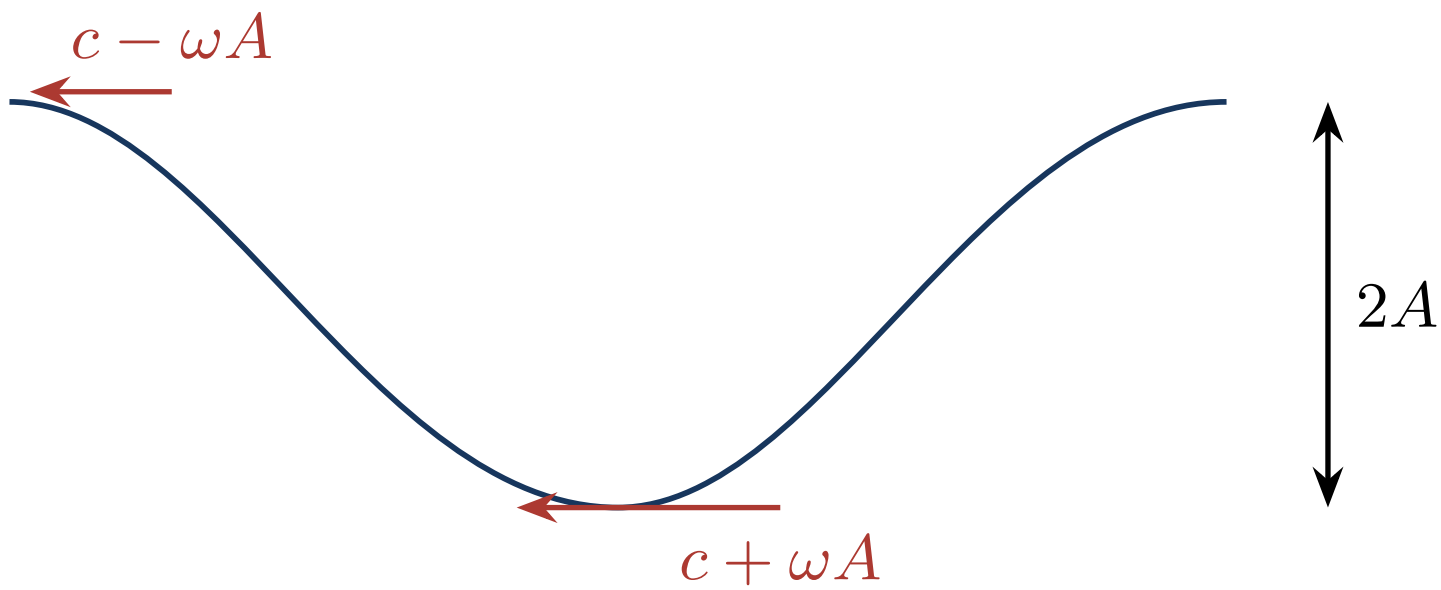


轨道速度与波形速度不同；示意中的振幅已放大



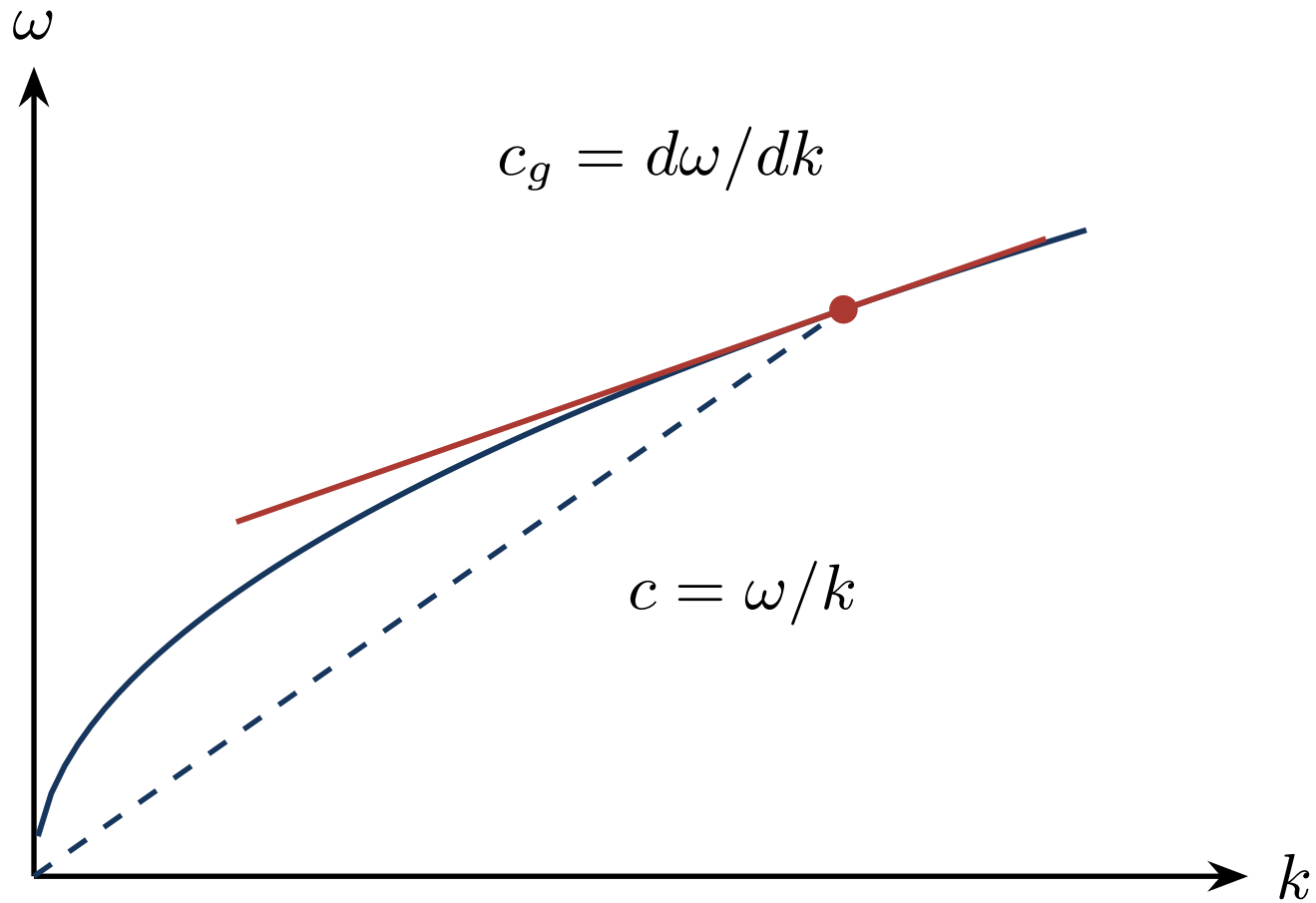
同一个参考系变换：$u_x = v_x - c$

自由面两处压强同为 p_0

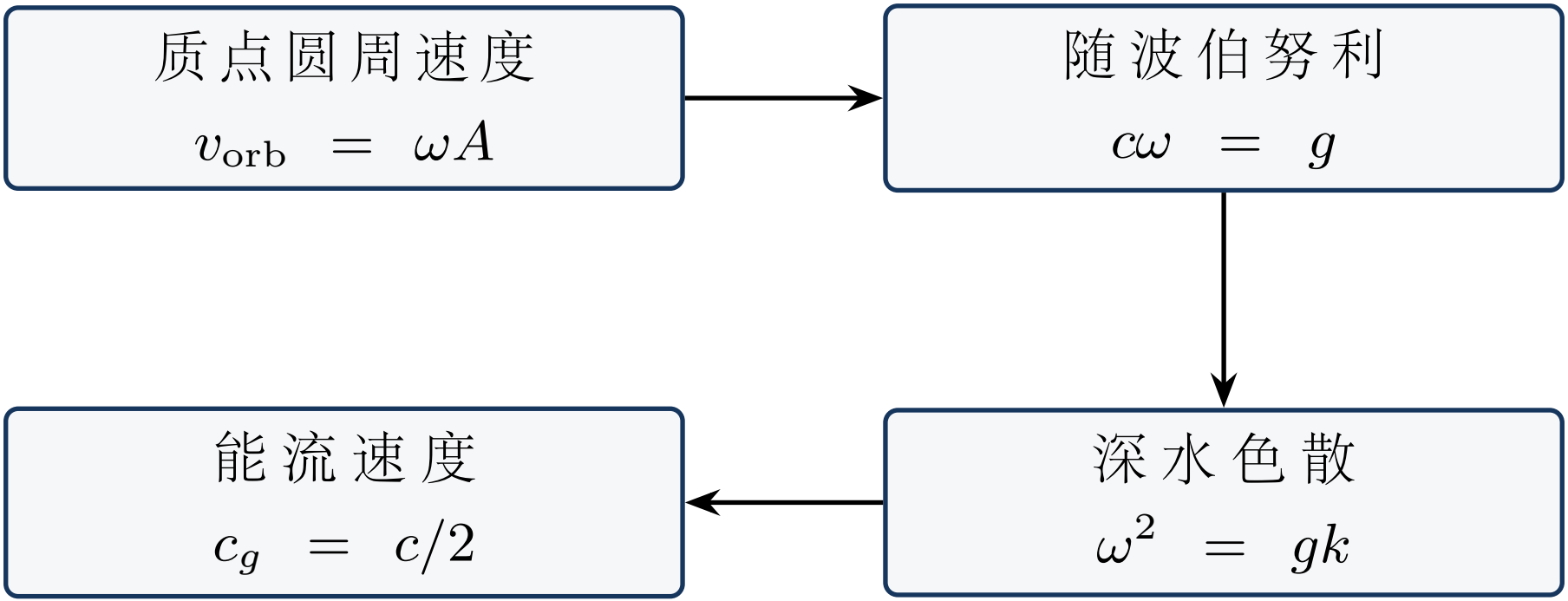


$$\frac{1}{2}[(c+\omega A)^2-(c-\omega A)^2] = 2gA$$

$$\text{故 } c\omega = g$$



$\omega = \sqrt{gk}$；深水群速度 $c_g = c/2$



深水 $kh_w \gg 1$ 与小斜率 $kA \ll 1$ 是两个独立条件